

Session-8-exercises.R

```
#####  
### Session 8 Exercises ###  
#####
```

```
### Class material
```

```
# Inspections
```

```
# Load the tidyverse and the food inspections dataset
```

```
library(tidyverse)
```

```
names <- c("ID", "DBAName", "AKAName", "License", "FacilityType", "Risk", "Address",  
           "City", "State", "ZIP", "InspectionDate", "InspectionType", "Results",  
           "Violations", "Latitude", "Longitude", "Location")
```

```
inspections <- read_csv('inspections.csv', col_names=names, skip=1)
```

```
# Create a new column called Regions that combines City and State
```

```
regional_inspections <- unite(inspections, Region, City, State, sep=", ",  
                               remove=FALSE)
```

```
# And take a look at the unique regions
```

```
unique(regional_inspections$Region)
```

```
## [1] "CHICAGO, IL"           "NA, IL"  
## [3] "ELK GROVE VILLAGE, IL" "SCHAUMBURG, IL"  
## [5] "Chicago, IL"          "CALUMET CITY, IL"  
## [7] "SKOKIE, IL"           "chicago, IL"  
## [9] "EVANSTON, IL"         "CCHICAGO, IL"  
## [11] "NILES NILES, IL"      "CHicago, IL"  
## [13] "ELMHURST, IL"        "NA, NA"  
## [15] "SCHILLER PARK, IL"    "OAK PARK, IL"  
## [17] "BRIDEVIEW, IL"       "CHESTNUT STREET, IL"  
## [19] "BEDFORD PARK, IL"    "BANNOCKBURNDEERFIELD, IL"  
## [21] "CHCICAGO, IL"        "BLOOMINGDALE, IL"  
## [23] "CICERO, IL"          "Norridge, IL"  
## [25] "CHICAGO, NA"         "CHARLES A HAYES, IL"  
## [27] "CHICAGOCHICAGO, IL"  "MAYWOOD, IL"  
## [29] "CHCHICAGO, IL"       "CHICAGOI, IL"  
## [31] "SUMMIT, IL"          "OOLYMPIA FIELDS, IL"  
## [33] "WESTMONT, IL"        "CHICAGO HEIGHTS, IL"  
## [35] "JUSTICE, IL"         "TINLEY PARK, IL"  
## [37] "LOMBARD, IL"         "EAST HAZEL CREST, IL"  
## [39] "COUNTRY CLUB HILLS, IL" "STREAMWOOD, IL"  
## [41] "BOLINGBROOK, IL"     "INACTIVE, IL"  
## [43] "BERWYN, IL"          "BURNHAM, IL"  
## [45] "DES PLAINES, IL"     "LAKE ZURICH, IL"  
## [47] "OLYMPIA FIELDS, IL"   "alsip, IL"  
## [49] "OAK LAWN, IL"        "BLUE ISLAND, IL"  
## [51] "GLENCOE, IL"         "FRANKFORT, IL"  
## [53] "NAPERVILLE, IL"    "BROADVIEW, IL"
```

```

## [55] "WORTH, IL"          "Maywood, IL"
## [57] "ALSIP, IL"          "EVERGREEN PARK, IL"

# We need to load stringr separately
library(stringr)

# Let's handle the uppercase/lowercase issues by converting everything to
uppercase
regional_inspections <- regional_inspections %>%
  mutate(Region=str_to_upper(Region))

# What were the results of that?
unique(regional_inspections$Region)

## [1] "CHICAGO, IL"          "NA, IL"
## [3] "ELK GROVE VILLAGE, IL" "SCHAUMBURG, IL"
## [5] "CALUMET CITY, IL"     "SKOKIE, IL"
## [7] "EVANSTON, IL"         "CCHICAGO, IL"
## [9] "NILES NILES, IL"      "ELMHURST, IL"
## [11] "NA, NA"               "SCHILLER PARK, IL"
## [13] "OAK PARK, IL"         "BRIDEVIEW, IL"
## [15] "CHESTNUT STREET, IL"  "BEDFORD PARK, IL"
## [17] "BANNOCKBURNDEERFIELD, IL" "CHCICAGO, IL"
## [19] "BLOOMINGDALE, IL"     "CICERO, IL"
## [21] "NORRIDGE, IL"         "CHICAGO, NA"
## [23] "CHARLES A HAYES, IL"   "CHICAGOCHICAGO, IL"
## [25] "MAYWOOD, IL"          "CHCHICAGO, IL"
## [27] "CHICAGOI, IL"         "SUMMIT, IL"
## [29] "OOLYMPIA FIELDS, IL"   "WESTMONT, IL"
## [31] "CHICAGO HEIGHTS, IL"   "JUSTICE, IL"
## [33] "TINLEY PARK, IL"       "LOMBARD, IL"
## [35] "EAST HAZEL CREST, IL"  "COUNTRY CLUB HILLS, IL"
## [37] "STREAMWOOD, IL"       "BOLINGBROOK, IL"
## [39] "INACTIVE, IL"         "BERWYN, IL"
## [41] "BURNHAM, IL"          "DES PLAINES, IL"
## [43] "LAKE ZURICH, IL"      "OLYMPIA FIELDS, IL"
## [45] "ALSIP, IL"            "OAK LAWN, IL"
## [47] "BLUE ISLAND, IL"      "GLENCOE, IL"
## [49] "FRANKFORT, IL"        "NAPERVILLE, IL"
## [51] "BROADVIEW, IL"        "WORTH, IL"
## [53] "EVERGREEN PARK, IL"

# Let's take care of a few misspellings of Chicago
badchicagos <- c('CCHICAGO, IL', 'CHCICAGO, IL', 'CHICAGOCHICAGO, IL',
  'CHCHICAGO, IL', 'CHICAGOI, IL')

regional_inspections <- regional_inspections %>%
  mutate(Region=ifelse(Region %in% badchicagos, 'CHICAGO, IL', Region))

# And see what's left
unique(regional_inspections$Region)

```

```
## [1] "CHICAGO, IL"      "NA, IL"
## [3] "ELK GROVE VILLAGE, IL" "SCHAUMBURG, IL"
## [5] "CALUMET CITY, IL"    "SKOKIE, IL"
## [7] "EVANSTON, IL"        "NILES NILES, IL"
## [9] "ELMHURST, IL"        "NA, NA"
## [11] "SCHILLER PARK, IL"   "OAK PARK, IL"
## [13] "BRIDGEVIEW, IL"      "CHESTNUT STREET, IL"
## [15] "BEDFORD PARK, IL"    "BANNOCKBURNDEERFIELD, IL"
## [17] "BLOOMINGDALE, IL"    "CICERO, IL"
## [19] "NORRIDGE, IL"        "CHICAGO, NA"
## [21] "CHARLES A HAYES, IL" "MAYWOOD, IL"
## [23] "SUMMIT, IL"          "OOLYMPIA FIELDS, IL"
## [25] "WESTMONT, IL"        "CHICAGO HEIGHTS, IL"
## [27] "JUSTICE, IL"         "TINLEY PARK, IL"
## [29] "LOMBARD, IL"         "EAST HAZEL CREST, IL"
## [31] "COUNTRY CLUB HILLS, IL" "STREAMWOOD, IL"
## [33] "BOLINGBROOK, IL"     "INACTIVE, IL"
## [35] "BERWYN, IL"          "BURNHAM, IL"
## [37] "DES PLAINES, IL"     "LAKE ZURICH, IL"
## [39] "OLYMPIA FIELDS, IL"   "ALSIP, IL"
## [41] "OAK LAWN, IL"        "BLUE ISLAND, IL"
## [43] "GLENCOE, IL"         "FRANKFORT, IL"
## [45] "NAPERVILLE, IL"    "BROADVIEW, IL"
## [47] "WORTH, IL"          "EVERGREEN PARK, IL"
```

There are some "CHICAGO, NA" values that we can clearly correct to "CHICAGO, IL"

```
regional_inspections <- regional_inspections %>%
  mutate(Region=ifelse(Region=='CHICAGO, NA', 'CHICAGO, IL', Region))
```

But we don't know what to do with "NA, IL", "NA, NA", or "INACTIVE, IL"
so let's set those to missing values

```
nachicagos <- c('NA, IL', 'NA, NA', 'INACTIVE, IL')
```

```
regional_inspections <- regional_inspections %>%
  mutate(Region=ifelse(Region %in% nachicagos, NA, Region))
```

How did we do?

```
unique(regional_inspections$Region)
```

```
## [1] "CHICAGO, IL"      NA
## [3] "ELK GROVE VILLAGE, IL" "SCHAUMBURG, IL"
## [5] "CALUMET CITY, IL"    "SKOKIE, IL"
## [7] "EVANSTON, IL"        "NILES NILES, IL"
## [9] "ELMHURST, IL"        "SCHILLER PARK, IL"
## [11] "OAK PARK, IL"        "BRIDGEVIEW, IL"
## [13] "CHESTNUT STREET, IL" "BEDFORD PARK, IL"
## [15] "BANNOCKBURNDEERFIELD, IL" "BLOOMINGDALE, IL"
## [17] "CICERO, IL"          "NORRIDGE, IL"
## [19] "CHARLES A HAYES, IL" "MAYWOOD, IL"
## [21] "SUMMIT, IL"          "OOLYMPIA FIELDS, IL"
## [23] "WESTMONT, IL"        "CHICAGO HEIGHTS, IL"
## [25] "JUSTICE, IL"         "TINLEY PARK, IL"
```

```
## [27] "LOMBARD, IL" "EAST HAZEL CREST, IL"
## [29] "COUNTRY CLUB HILLS, IL" "STREAMWOOD, IL"
## [31] "BOLINGBROOK, IL" "BERWYN, IL"
## [33] "BURNHAM, IL" "DES PLAINES, IL"
## [35] "LAKE ZURICH, IL" "OLYMPIA FIELDS, IL"
## [37] "ALSIP, IL" "OAK LAWN, IL"
## [39] "BLUE ISLAND, IL" "GLENCOE, IL"
## [41] "FRANKFORT, IL" "NAPERVILLE, IL"
## [43] "BROADVIEW, IL" "WORTH, IL"
## [45] "EVERGREEN PARK, IL"
```

McDo Forever

Check most inspected restaurants

```
inspections %>%
```

```
  group_by(DBAName) %>%
```

```
  summarize(inspections=n()) %>%
```

```
  arrange(desc(inspections))
```

```
## # A tibble: 24,032 × 2
```

DBAName	inspections
<chr>	<int>
1 SUBWAY	2020
2 DUNKIN DONUTS	1023
3 MCDONALD'S	458
4 7-ELEVEN	363
5 MCDONALDS	260
6 CHIPOTLE MEXICAN GRILL	243
7 POTBELLY SANDWICH WORKS LLC	212
8 CORNER BAKERY CAFE	181
9 SPORTSERVICE SOLDIER FIELD	176
10 POTBELLY SANDWICH WORKS	166

```
## # i 24,022 more rows
```

```
inspections <- inspections %>%
```

```
  mutate(Mcdo=str_detect(inspections$DBAName, pattern="MCDO"))
```

```
sum(inspections$Mcdo)
```

```
## [1] 1163
```

#alternatively

```
sum(str_detect(inspections$DBAName, pattern="MCDO"))
```

```
## [1] 1163
```

#The list of wrongly spelled McDo

```
wrongMcDo <- unique(str_subset(inspections$DBAName, pattern="MCDO"))
wrongMcDo
```

```
## [1] "MCDONALD"
## [2] "MCDONALDS"
## [3] "MCDONALD'S #20104"
## [4] "MCDONALD'S"
## [5] "MCDONALDS #29307"
## [6] "MCDONALD'S RESTAURANTS"
```

```
## [7] "MCDONALD'S RESTAURANT"
## [8] "MCDONALD'S #4305"
## [9] "MCDONALD'S CORPORATION"
## [10] "MCDONALD'S #5813"
## [11] "MCDONALDS RESTAURANT"
## [12] "MCDONALD'S #26364"
## [13] "MCDONALD'S #490"
## [14] "MCDONALD'S 26371"
## [15] "MCDONALDS #12003"
## [16] "MCDONALD'S #6491"
## [17] "MCDONALDS RESTAURANT # 14173"
## [18] "MCDONALDS #27672"
## [19] "MCDONALD'S REST. OF ILLINOIS, INC"
## [20] "MCDONALDS #5471"
## [21] "MCDONALD' S # 5618"
## [22] "MCDONALDS #4655"
## [23] "MCDONALD'S #6337"
## [24] "MCDONALD'S RESTAURANT OF ILLINOIS, INC"
## [25] "MCDONALD'S #11290"
## [26] "MCDONALDS 812"
## [27] "MCDONALD'S #6310"
## [28] "MCDONALDS #7069"
## [29] "MCDONALDS'S RESTAURANTS"
## [30] "MCDONALD'S REST. OF ILL, INC"
## [31] "MCDONALDS #7463"
## [32] "ROCK & ROLL MCDONALDS"
## [33] "MCDONALD'S #5246"
## [34] "ERNEST CORP / MCDONALD'S"
## [35] "MCDONALDS #490"
## [36] "SARAH MCDONALD STEELE"
## [37] "MCDONALD' S RESTAURANT"
```

Find alternate spellings of McDonalds

```
inspections %>%
  filter(grepl("McDo", DBAName, ignore.case=TRUE)) %>%
  select(DBAName) %>%
  unique() %>%
  View()
```

```
inspections %>%
  filter(grepl("McDo", DBAName, ignore.case=TRUE)) %>%
  filter(DBAName != 'SARAH MCDONALD STEELE') %>%
  select(DBAName) %>%
  unique() %>%
  View()
```

Create a vector of those alternate spellings

```
alternates <- inspections %>%
  filter(grepl("McDo", DBAName, ignore.case=TRUE)) %>%
  filter(DBAName != 'SARAH MCDONALD STEELE') %>%
  select(DBAName) %>%
  unique() %>%
  pull(DBAName)
```

alternates

```
## [1] "MCDONALD"
## [2] "MCDONALDS"
## [3] "MCDONALD'S #20104"
## [4] "MCDONALD'S"
## [5] "MCDONALDS #29307"
## [6] "McDONALD'S"
## [7] "MCDONALD'S RESTAURANTS"
## [8] "McDonald's Restaurant"
## [9] "McDonald's"
## [10] "McDONALDS"
## [11] "McDonalds"
## [12] "MCDONALD'S RESTAURANT"
## [13] "MCDONALD'S #4305"
## [14] "MCDONALD'S CORPORATION"
## [15] "MCDONALD'S #5813"
## [16] "McDowell"
## [17] "McDonald's #4308"
## [18] "McDONALD'S # 17277"
## [19] "McDONALD'S # 12785"
## [20] "McDONALD'S # 17274"
## [21] "McDONALDS # 17278"
## [22] "MCDONALDS RESTAURANT"
## [23] "MCDONALD'S #26364"
## [24] "McDONALD'S # 22821"
## [25] "MCDONALD'S #490"
## [26] "MCDONALD'S 26371"
## [27] "MCDONALDS #12003"
## [28] "MCDONALD'S #6491"
## [29] "MCDONALDS RESTAURANT # 14173"
## [30] "MCDONALDS #27672"
## [31] "McDONALD'S STORE #4061"
## [32] "MCDONALD'S REST. OF ILLINOIS, INC"
## [33] "MCDONALDS #5471"
## [34] "MCDONALD' S # 5618"
## [35] "McDonalds Restaurant"
## [36] "MCDONALDS #4655"
## [37] "MCDONALD'S #6337"
## [38] "MCDONALD'S RESTAURANT OF ILLINOIS, INC"
## [39] "MCDONALD'S #11290"
## [40] "MCDONALDS 812"
## [41] "McDonalds #6337"
## [42] "MCDONALD'S #6310"
## [43] "MCDONALDS #7069"
## [44] "MCDONALDS'S RESTAURANTS"
## [45] "MCDONALD'S REST. OF ILL, INC"
## [46] "MCDONALDS #7463"
## [47] "ROCK & ROLL MCDONALDS"
## [48] "MCDONALD'S #5246"
## [49] "ERNEST CORP / MCDONALD'S"
## [50] "MCDONALDS #490"
## [51] "MCDONALD' S RESTAURANT"
## [52] "mcdonalds"
```

```
# Replace them all with MCDONALDS
inspections <- inspections %>%
  mutate(DBAName=ifelse(DBAName %in% alternates, 'MCDONALDS', DBAName))
```

```
# Check most inspected restaurants again
```

```
inspections %>%
  group_by(DBAName) %>%
  summarize(inspections=n()) %>%
  arrange(desc(inspections))
```

```
## # A tibble: 23,981 × 2
```

##	DBAName	inspections
##	<chr>	<int>
## 1	SUBWAY	2020
## 2	MCDONALDS	1457
## 3	DUNKIN DONUTS	1023
## 4	7-ELEVEN	363
## 5	CHIPOTLE MEXICAN GRILL	243
## 6	POTBELLY SANDWICH WORKS LLC	212
## 7	CORNER BAKERY CAFE	181
## 8	SPORTSERVICE SOLDIER FIELD	176
## 9	POTBELLY SANDWICH WORKS	166
## 10	DOMINO'S PIZZA	162
## # i	23,971 more rows	